

A New Feature Extraction Method Based on Micro-Doppler Signature of Ground Moving Targets

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Abstract—Since some discriminative features are required for ground moving targets classification, we propose a new feature extraction method based on micro-Doppler signature. Firstly, the time-frequency analysis of measured data indicates that the time-frequency spectrograms of the three kinds of ground moving targets, i.e., single walking person, two people walking and a moving wheeled vehicle, are discriminative. Then, a 3-dimensional time-frequency feature vector is extracted from the time-frequency spectrograms to depict these differences. At last, a Support Vector Machine (SVM) classifier is trained with the proposed 3-dimensional feature vector. The classification accuracy to categorize ground moving targets into the three kinds on the measured data is found to be over 96%, which demonstrates the good discriminative ability of the proposed micro-Doppler feature.

Keywords—Micro-Doppler, Time-Frequency Analysis, Feature Extraction, Radar Target Classification

I. INTRODUCTION

It is well known that moving targets will cause a Doppler frequency shift into the radar returned signal, referred to as Doppler Effect [1]. If there are structural components rotating or oscillating on the target, these moving components will induce additional frequency modulations around the center frequency of the target, which is referred to as micro-Doppler effect [2]. Since different kinds of targets usually induce different types of micro-Doppler modulations in the low resolution radar returns, some discriminative features may be extracted from the mentioned modulations to describe the different targets. As a consequence, there have been lots of researches applying the micro-Doppler effect to the radar target detection and classification field [3-5]. After analyzing the related signatures for pedestrians and vehicles, [6,7] extract the corresponding micro-Doppler features for classification. In [8,9], the micro-Doppler information contained in the time-frequency spectrograms of different human gait motions is utilized for human activity classification. [10] proposes a hierarchical micro-Doppler feature extraction method to categorize moving vehicles into two kinds, i.e., wheeled vehicle and tracked vehicle.

In this paper, we analyze the signal models and measured data from a low resolution radar to find the differences between the micro-Doppler signatures from the three kinds of ground moving targets, i.e., single walking person, two people walking and a moving wheeled vehicle. Then we define a

novel 3-dimensional micro-Doppler feature vector extracted from the time-frequency spectrograms to represent the micro-motion characteristics of three kinds of ground moving targets. Additionally, we utilize the ratio of the between-class distance to the within-class distance which is defined based on Fisher Discriminant Analysis (FDA)[11] and SVM classifier to demonstrate that the proposed features have a higher degree of discriminative ability than some existing features.

II. MICRO-DOPPLER SIGNATURE OF MEASURED DATA

Our measured data used in this paper are collected from the three kinds of ground moving targets. The low resolution radar used in the experiment is a Ka band system with its Pulse Repetition Frequency (PRF) F_{PRF} being 3 KHz, its dwell time being 1.2 s, the pulse length being 3 μ s for each transmitted Linear Frequency Modulated (LFM) pulse and the bandwidth being 18 MHz. There are 120 acquisitions for each kind of considered target. Before micro-Doppler feature extraction, we use the CLEAN technique [12] to remove the ground clutter components in our original measured data.

When people walk, there are various types of simultaneous movements such as the swinging motions of the arms and legs, the lateral translation motion of the spine, et al. Since all the simultaneous movements are periodic, they induce periodic micro-Doppler modulations in the radar return. The instantaneous frequency of each periodic modulation may be approximately formulated as $\sum_{i=1}^M \sum_{j=1}^N A_{ij} \sin(2\pi f_{ij}t + \varphi_{ij})$, where M denotes the number of walking people, N denotes the number of movements existed of single person, φ_{ij} denotes the initial phase of the j th movement of the i th person, A_{ij} and f_{ij} denote the amplitude and the frequency of the j th movement of the i th person respectively. We empirically know that A_{ij} and f_{ij} are related to the radius of different body motions and the walking speed.

Since the lengths of arms and legs are much larger than the motion radiiuses of the torso components, the micro-Doppler modulations induced by the swinging arms and legs will capture a much wider frequency bandwidth. The time-frequency spectrograms of measured data depicted in Fig. 1(a) and Fig. 1(b) are in accordance with the above analysis.

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The micro-Doppler modulations induced by a moving wheeled vehicle are quite different from those of walking people. When a wheeled vehicle is moving, micro-Doppler components and the bulk-motion component of a moving vehicle does not vary periodically with time. In addition, the bulk component can be regarded as an ensemble due to the rigid structure characteristics of a vehicle. Therefore the modulation induced by the bulk component should be represented as $A \exp(j4\pi vt/\lambda)$ and it captures most of the energy of the radar returns from a moving wheeled vehicle, where v denotes the speed of the translation motion and A denotes the scattering coefficient of the bulk component, as shown in Fig. 1(c).

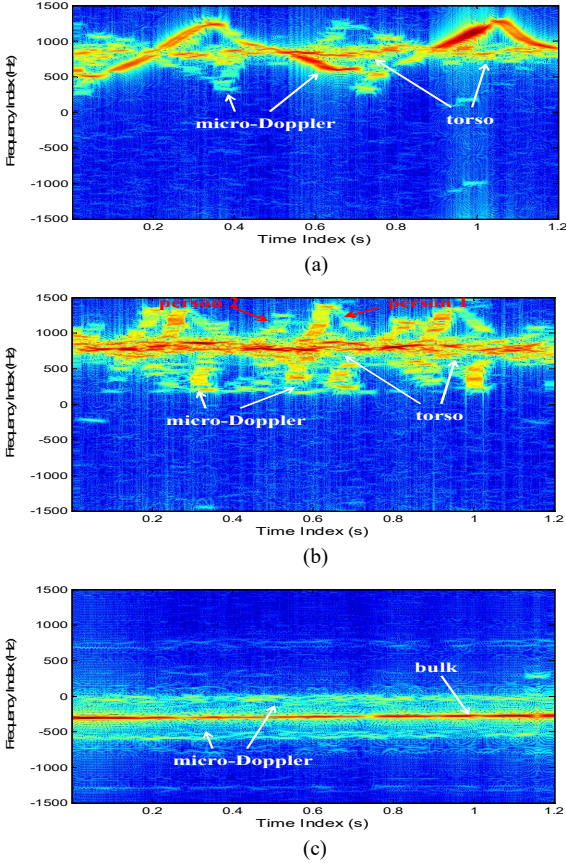


Fig. 1. Time-frequency spectrograms of some measured data. (a) single walking person; (b) two people walking; (c) a moving wheeled vehicle.

III. TIME-FREQUENCY FEATURE EXTRACTION

Fig. 1 highlights that there are some differences among the micro-Doppler modulations of the considered targets. We use the following three features to depict the differences of the micro-Doppler signatures reflected in the time-frequency spectrograms of the considered targets.

Symbol Notation:

1) $\tilde{S} = [\tilde{S}_{f,t}]_{f=1, t=1}^{N_F, N_T}$ denotes the time-frequency spectrogram, which is the modulus matrix of STFT of the acquisition. Here f is the row index corresponding to the frequency dimension in STFT, t is the column index corresponding to the time dimension in STFT, N_F represents the length of the Fast Discrete Fourier Transformation (FDFT) for each short time signal and N_T represents the number of samples in the time dimension in STFT. In our experiments,

$N_F = 3000$ and $N_T = T \cdot F_{PRF} = 3600$.
 $\bar{S} = \frac{1}{N_F \times N_T} \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} \tilde{S}_{f,t}$ denotes the mean of the time-frequency spectrogram.

$$2) \quad \mathbf{E} = [E_f]_{f=1}^{N_F} = \left[\frac{1}{N_T} \sum_{t=1}^{N_T} S_{f,t} \right]_{f=1}^{N_F} = \frac{1}{N_T} \left[\sum_{t=1}^{N_T} S_{f,t} \right]_{f=1}^{N_F}$$

denotes the mean vector consisting of mean value of each row in the normalized time-frequency spectrogram with

$$S_{f,t} = \tilde{S}_{f,t} / \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} \tilde{S}_{f,t} \quad ; \text{since} \quad \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} S_{f,t} = 1,$$

$\bar{E} = \frac{1}{N_F} \sum_{f=1}^{N_F} E_f = \frac{1}{N_F} \frac{1}{N_T} \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} S_{f,t} = \frac{1}{N_F \cdot N_T}$ denotes the mean of $\mathbf{E}$; $f_{\max} = \arg \max \mathbf{E}$ denotes the index corresponding to the maximum value in $\mathbf{E}$; f_{upper} denotes the index that satisfies $E_{f_{\text{upper}}} \geq \bar{E}$, $E_{f_{\text{upper}}+1} < \bar{E}$ and nearest to $f_{\max}$ within the range $[f_{\max}, N_F]$, f_{lower} denotes the index that satisfies $E_{f_{\text{lower}}} \geq \bar{E}$, $E_{f_{\text{lower}}-1} < \bar{E}$ and nearest to $f_{\max}$ within the range $[1, f_{\max}]$. The above definitions are illustrated in the left subfigures in Fig. 2.

3) $\mathbf{G} = [G_t]_{t=1}^{N_T}$ with $G_t = \arg \max_f \left([\tilde{S}_{f,t}]_{f=1}^{N_F} \right)$, where G_t denotes the row index of the largest value in the t th column of $\tilde{S}$; $\bar{G} = \frac{1}{N_T} \sum_{t=1}^{N_T} G_t$ denotes the mean of $\mathbf{G}$. The above definitions are illustrated in the bottom subfigures in Fig. 2.

Feature 1: the variance of the time-frequency spectrogram.

$$\mathcal{F}(1) = \frac{1}{N_F \times N_T} \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} (\tilde{S}_{f,t} - \bar{S})^2 \quad (1)$$

According to the property of variance, the value of $\mathcal{F}(1)$ is determined by the fluctuation degree of the time-frequency plane. For the time-frequency spectrogram of a moving vehicle shown in the middle subfigure of Fig. 2(c), the whole time-frequency plane is smooth beside the range captured by the bulk component. Thus the corresponding value of $\mathcal{F}(1)$ of a moving wheeled vehicle is the largest. Since there are more abundant micro-Doppler modulations in the time-frequency spectrograms of two people walking than that of single walking person, the value of $\mathcal{F}(1)$ of two people walking is smaller than that of single walking person.

Feature 2: the bandwidth of the targets' Doppler modulations in the time-frequency spectrogram.

$$\mathcal{F}(2) = f_{\text{upper}} - f_{\text{lower}} \quad (2)$$

As shown in the left subfigures in Fig. 2, the bandwidth of the targets' Doppler modulations for the single walking person is much wider than that of a moving vehicle; while two people walking have a wider induced modulation bandwidth than a single walking person due to more complicated micro-Doppler modulation.

Feature 3: the variance of the frequencies corresponding to the largest values in each column of the time-frequency spectrogram.

$$\mathbb{F}(3) = \frac{1}{N_T} \sum_{t=1}^{N_T} (G_t - \bar{G})^2 \quad (3)$$

Since the energy of the bulk motion is much larger than that of the micro-Doppler motion for a moving vehicle, the instantaneous frequencies corresponding to the largest values in each column of the time-frequency spectrogram always correspond to the bulk motion component. As shown in the bottom subfigure of Fig. 2(c), when a vehicle is moving, the corresponding frequency induced by the moving bulk is close to a constant within the short dwell time. However, for the human target, frequency modulation induced by the torso motion corresponds to a much wider frequency range than that induced by the bulk motion, and the energy of the torso motion is more spread than that of the bulk motion as illustrated in the middle subfigures of Fig. 2. From the bottom subfigures of Fig. 2(a) and (b) we can see that the curves of $\mathbf{G}$ for single walking person and two people walking both vary with time, therefore, their values of $\mathbb{F}(3)$ are much larger than that of a moving vehicle. For a single walking person, the energy of the micro-Doppler component is close to that of the torso component. However, for two people walking, the energy of the torso component is often larger than that of the micro-Doppler component since the arms and legs are easily sheltered. As shown in the bottom subfigures of Fig. 2(a) and (b), the curve of $\mathbf{G}$ for single walking person fluctuates more severely than that of two people walking. Thus the value of this feature for single walking person is larger than that of two people walking.

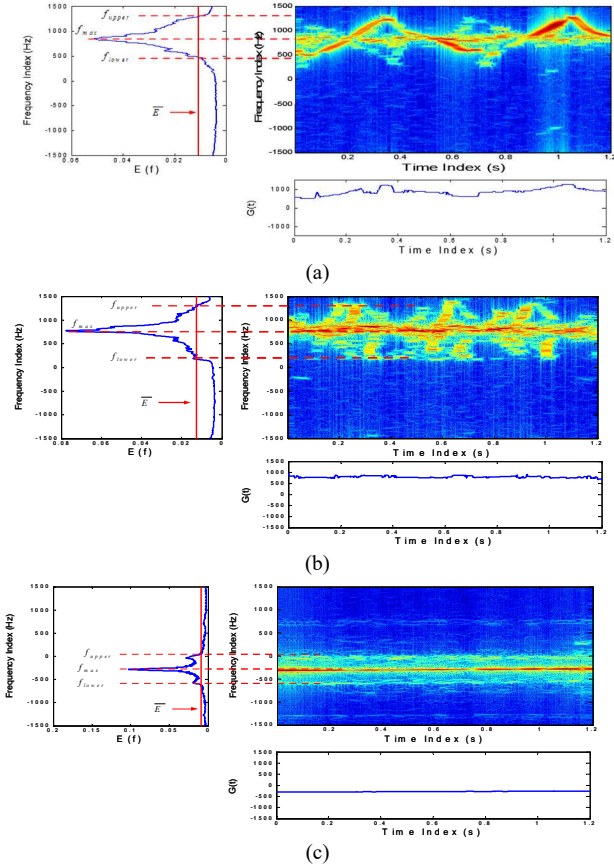


Fig. 2. Feature extraction examples for the measured data from the three kinds of considered targets, where the time-frequency spectrogram, i.e., $\tilde{\mathbf{S}}$, is shown in the middle subfigure, the mean vector consisting of mean value of each row in the normalized time-frequency spectrogram, i.e., $\mathbf{E}$, is shown in the left subfigure, and the frequencies corresponding to the largest values in each column of the time-frequency spectrogram, i.e., $\mathbf{G}$, is shown in the bottom subfigure. (a) single walking person; (b) two people walking; (c) a moving wheeled vehicle.

IV. EXPERIMENT RESULTS OF MEASURED DATA

In this section, we utilize the measured data to evaluate the classification performance of the proposed 3-dimensional feature vector. First, we divide the measured data into two sets, i.e., training data set and test data set. Then the 3-dimensional micro-Doppler feature is extracted from the time-frequency spectrogram of each acquisition in the training data set and test data set. In the training stage, we use the feature data of training acquisitions to train a SVM classifier; while in the test stage the trained classifier is utilized to categorize each feature data from test acquisitions into corresponding category.

A. Micro-Doppler Feature Analysis

In order to evaluate the discriminative ability of the proposed 3-dimensional feature vector, we utilize the FDA which can be represented as

$$FDA = S_B / S_W = \frac{\sum_{k=1}^K \sum_{j>k}^K (m_k - m_j)^2}{\sum_{x \in C_k} (x - m_k)^2} \quad (4)$$

Where S_B and S_W denote the between-class distance and the within-class distance respectively, $x \in C_k$ denotes the feature vector corresponding to the k th class C_k , m_k is the mean vector of the k th class, i.e., $m_k = \frac{1}{N_k} \sum_{x \in C_k} x$ with N_k denoting the number of vectors in the k th class. The small within-class distance means the feature vectors are close to each other in the category and the large between-class distance means that the mean of feature vector of each category is far away from others. An ideal feature vector will achieve the large between-class distance and small within-class distance, i.e. the large FDA.

Fig. 3 gives the values of FDA for the feature vectors obtained from the training set (for each training feature, the corresponding category is known). $\mathbb{F}(1) \sim \mathbb{F}(3)$ are the proposed 3-dimensional feature described in Section III. We also calculate the between-class distance and within-class distance for $\mathbb{F}(4) \sim \mathbb{F}(6)$ described in paper [7] and $\mathbb{F}(7) \sim \mathbb{F}(10)$ described in paper [9] for comparison. Based on the results in Fig. 3 we can see that the proposed $\mathbb{F}(1) \sim \mathbb{F}(3)$ are more discriminative than $\mathbb{F}(4) \sim \mathbb{F}(10)$ in the feature domain.

B. Classification Results

The features extracted in the training stage are utilized to train a SVM which is then used to determine the category of each test acquisition. In our experiment, we choose the Gaussian function as the kernel of SVM. Table I, Table II and Table III show the confusion matrixes of the classification accuracies for each target category and the

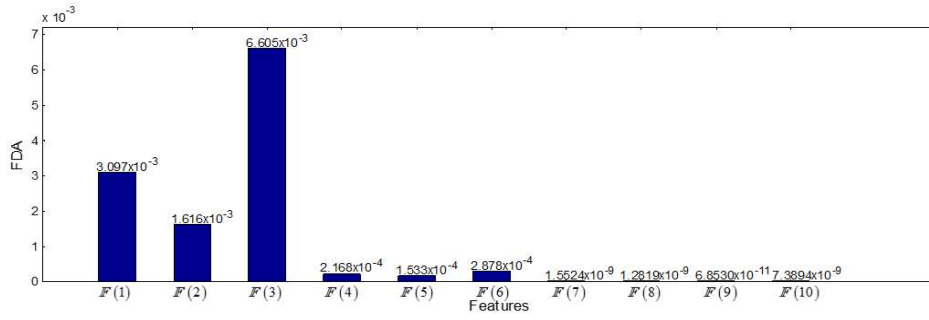


Fig. 3. Values of FDA for different features

average classification rate. The average classification rate is calculated using the number of test acquisitions categorized into its right class divided by the totally number of test acquisition. From the results shown in Table I, we can see that our proposed method can categorize test samples into the right class with the accuracy over 96%. However, as shown in Table II, the 3-dimensional feature vector proposed in [7] wrongly recognize the test acquisitions of the single walking person to be those of the moving vehicle with the rate over 10%, and vice versa. As shown in Table III, the 4-dimensional feature vector proposed in [9] tends to wrongly recognize the test acquisitions of the single walking person to be those of the moving vehicle with the rate over 30%. Therefore, comparing the results in Table I, II, III, our proposed 3-dimensional feature vector achieves a much higher classification accuracy than the 3-dimensional feature vector of [7] and the 4-dimensional feature vector of [9].

V. CONCLUSION

In this paper, we propose a novel feature extraction method based on the micro-Doppler signatures in the time-frequency spectrograms to categorize ground moving targets into three kinds, *i.e.*, single walking person, two people walking and a moving wheeled vehicle. Experiment results based on measured data show the proposed micro-Doppler features have a higher discriminative ability and higher classification accuracy than some existing features.

TABLE I. CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE PROPOSED 3-DIMENSIONAL FEATURE VECTOR (%)

	<i>Single</i>	<i>Two</i>	<i>Vehicle</i>	<i>Average</i>
Single	91.67	1.670	1.670	96.11
Two	1.670	98.33	0	
Vehicle	6.670	0	98.33	

TABLE II. CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE 3-DIMENSIONAL FEATURE VECTOR PROPOSED IN [7] (%)

	<i>Single</i>	<i>Two</i>	<i>Vehicle</i>	<i>Average</i>
Single	80.00	3.330	11.67	87.78
Two	8.330	95.00	0	
Vehicle	11.67	1.670	88.33	

TABLE III. CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE 4-DIMENSIONAL FEATURE VECTOR PROPOSED IN [9] (%)

	<i>Single</i>	<i>Two</i>	<i>Vehicle</i>	<i>Average</i>
Single	65.00	3.330	0	84.44
Two	5.00	88.33	0	
Vehicle	30.00	8.340	100	

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